

Tp2 PIG

Exercice 1: Exemple Word Count - Pig Script

1- En mode interactif (Grunt shell)

```

grunt> lines = LOAD '/user/cloudera/pig/poemes.txt' AS (line:chararray);
grunt> words = FOREACH lines GENERATE FLATTEN(TOKENIZE(line)) as word;
grunt> grouped = GROUP words BY word;
grunt> wordcount = FOREACH grouped GENERATE group, COUNT(words);
grunt> DUMP wordcount;
2024-02-06 03:58:57,261 [main] INFO org.apache.pig.tools.pigstats.ScriptState -
Pig features used in the script: GROUP_BY
2024-02-06 03:58:57,329 [main] INFO org.apache.pig.newplan.logical.optimizer.Lo
gicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, DuplicateFor
EachColumnRewrite, GroupByConstParallelSetter, ImplicitSplitInserter, LimitOptim
izer, LoadTypeCastInserter, MergeFilter, MergeForEach, NewPartitionFilterOptimiz
er, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter],
RULES_DISABLED=[FilterLogicExpressionSimplifier, PartitionFilterOptimizer]}
[! ,9]
[. ,3]
[: ,11]
[; ,4]
[? ,2]
[Ce ,3]
[Et ,4]
[Je ,1]
[Le ,3]
[On ,1]
[Un ,1]
[au ,3]
[ce ,1]
[de ,16]
[du ,4]
[en ,2]

```

2- En mode Batch (script)

```
[cloudera@quickstart ~]$ touch wordcount.pig
[cloudera@quickstart ~]$ cat wordcount.pig
lines = LOAD '/user/cloudera/data-pig/poemes.txt' AS (line:chararray);
words = FOREACH lines GENERATE FLATTEN(TOKENIZE(line)) as word;
grouped = GROUP words BY word;
wordcount = FOREACH grouped GENERATE group, COUNT(words);
DUMP wordcount;
[cloudera@quickstart ~]$
```

```
---
grunt> exec wordcount.pig
2024-02-11 04:49:12,821 [main] INFO org.apache.hadoop.conf.Configuration:
d. Instead, use fs.defaultFS
2024-02-11 04:49:12,821 [main] INFO org.apache.hadoop.conf.Configuration:
ated. Instead, use mapreduce.jobtracker.address

Input(s):
Successfully read 61 records (2579 bytes) from: "/user/cloudera/pig/poemes.txt"

Output(s):
Successfully stored 236 records (2562 bytes) in: "hdfs://quickstart.cloudera:802
0/tmp/temp1992653945/tmp-93977161"

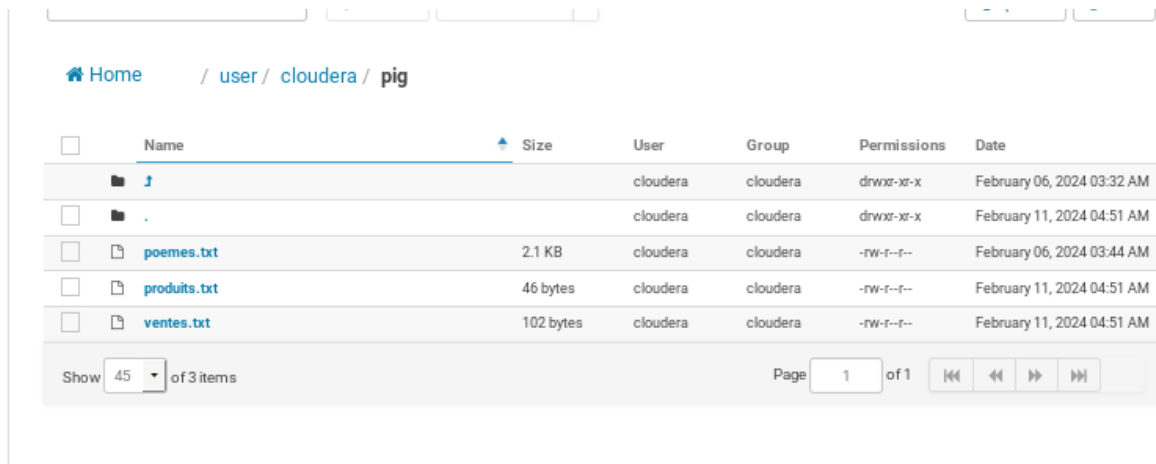
Counters:
Total records written : 236
Total bytes written : 2562
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1707166712860_0006

2024-02-11 04:46:53,138 [main] INFO org.apache.pig.backend.hadoop.executionengi
ne.mapReduceLayer.MapReduceLauncher - Success!
2024-02-11 04:46:53,138 [main] INFO org.apache.hadoop.conf.Configuration.deprec
ation - fs.default.name is deprecated. Instead, use fs.defaultFS
2024-02-11 04:46:53,138 [main] INFO org.apache.hadoop.conf.Configuration.deprec
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr
ess
2024-02-11 04:46:53,139 [main] INFO org.apache.pig.data.SchemaTupleBackend - Ke
y [pig.schematuple] was not set... will not generate code.
2024-02-11 04:46:53,177 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileI
nputFormat - Total input paths to process : 1
2024-02-11 04:46:53,177 [main] INFO org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(!,9)
(.,3)
(:,11)
(;;,4)
(?,2)
(Ce,3)
(Et,4)
(Je,1)
(Le,3)
(On,1)
(Un,1)
(au,3)
(ce,1)
(de,16)
```

Exercice 2 : Création de schéma, lecture et écriture de données avec Pig

Question : Copier le dossier data-pig sous HDFS (Utiliser HUE par exemple)



☐	Name	Size	User	Group	Permissions	Date
☐	.		cloudera	cloudera	drwxr-xr-x	February 06, 2024 03:32 AM
☐	poemes.txt	2.1 KB	cloudera	cloudera	-rw-r--r--	February 06, 2024 03:44 AM
☐	produits.txt	46 bytes	cloudera	cloudera	-rw-r--r--	February 11, 2024 04:51 AM
☐	ventes.txt	102 bytes	cloudera	cloudera	-rw-r--r--	February 11, 2024 04:51 AM

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Question : Afficher le contenu de la relation A sur la console

```
2024-02-13 03:51:37,600 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Failed!
2024-02-13 03:51:37,601 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR : 66: Unable to open iterator for alias A
Details at logfile: /home/cloudera/pig/1707823195094.log
grunt> A = LOAD '/user/cloudera/pig/produits.txt';
grunt> describe A;
Schema for A unknown.
grunt> DUMP A;
2024-02-13 03:51:59,247 [main] INFO org.apache.pig.tools.pigstats.ScriptState
Pig features used in the script: UNKNOWN
2024-02-13 03:51:59,247 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, DuplicateForEachColumnRewrite, GroupByConstParallelSetter, ImplicitSplitInserter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NewPartitionFilterOptimizer, PushDownForEachFilter, PushDownFilter, SplitFilter, StoreTypeCastOptimizer]}
---
2024-02-13 03:52:24,332 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to process : 1
2024-02-13 03:52:24,332 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(10,iphone,1000)
(20,samsung,2000)
(30,nokia,3000)
```

Question : Charger la relation A dans un dossier « produits » du HDFS

```
grunt> STORE A INTO '/user/cloudera/pig/produits';
2024-02-13 03:56:16,552 [main] INFO org.apache.hadoop.conf.Configuration - fs.default.name is deprecated. Instead, use fs.defaultFS
2024-02-13 03:56:16,552 [main] INFO org.apache.hadoop.conf.Configuration - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker
```

Home / user / cloudera / pig / produits

☐	Name	Size	User	Group	Permissions	Date
☐	f		cloudera	cloudera	drwxr-xr-x	February 13, 2024 03:56 AM
☐	.		cloudera	cloudera	drwxr-xr-x	February 13, 2024 03:56 AM
☐	_SUCCESS	0 bytes	cloudera	cloudera	-rw-r--r--	February 13, 2024 03:56 AM
☐	part-m-00000	45 bytes	cloudera	cloudera	-rw-r--r--	February 13, 2024 03:56 AM

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- Définition de schéma avec les noms de champs et les types de données.

Question : Charger la relation A dans un dossier « produits-copy » du HDF en remplaçant le délimiteur « , » par « | ».

```
grunt> A = LOAD '/user/cloudera/pig/produits.txt' USING PigStorage('|') AS (id:int, product_name:chararray, price:int);
grunt> describe A;
A: {id: int,product_name: chararray,price: int}
grunt> DUMP A;
2024-02-13 04:03:28,330 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: UNKNOWN
2024-02-13 04:03:28,331 [main] INFO org.apache.pig.newplan.logical.optimizer.Lo
```

2.2 Accès aux champs :

Questions :

- 1- Stocker dans une relation B tous les champs id des produits
- 2- Stocker dans une relation C tous les champs « nom et prix » des produits

```
grunt> A = LOAD '/user/cloudera/pig/produits.txt' USING PigStorage('|') AS (id:int, product_name:chararray, price:int);
grunt> B = FOREACH A GENERATE id;
C = FOREACH A GENERATE product_name, price;
DUMP B;
DUMP C;
2024-02-13 04:06:43,970 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: UNKNOWN
2024-02-13 04:06:43,972 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, DuplicateForEachColumnRewrite, GroupByConstParallelSetter, ImplicitSplitInserter, LimitOptimizer, LoadTypeCastInserter,
```

Exercice 3 : Comment filtrer les enregistrements – Script Pig

```
grunt> sales = LOAD '/user/cloudera/pig/ventes.txt' USING PigStorage(',') AS (product_id:
int, quantity:int, year:int);
grunt> filtered_products 1 = FILTER sales BY quantity >= 1000;
grunt> DUMP filtered_products 1;
2024-02-13 10:33:49,573 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig feat
ures used in the script: FILTER
2024-02-13 10:33:49,575 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlan
Optimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, DuplicateForEachColumnRewrite,
GroupByConstParallelSetter, ImplicitSplitInserter, LimitOptimizer, LoadTypeCastInserter,
MergeFilter, MergeForEach, NewPartitionFilterOptimizer, PushDownForEachFlatten, PushUpFi
lter, SplitFilter, StreamTypeCastInserter], RULES_DISABLED=[FilterLogicExpressionSimplifi
```

```
grunt> filtered_products 2 = FILTER sales BY quantity > 1000 AND year == 2001;
grunt> DUMP filtered_products 2;
2024-02-13 12:11:34,302 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig feat
res used in the script: FILTER
2024-02-13 12:11:34,306 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlan
ptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, DuplicateForEachColumnRewrite,
roupByConstParallelSetter, ImplicitSplitInserter, LimitOptimizer, LoadTypeCastInserter, M
rgeFilter, MergeForEach, NewPartitionFilterOptimizer, PushDownForEachFlatten, PushUpFilte
r, SplitFilter, StreamTypeCastInserter], RULES_DISABLED=[FilterLogicExpressionSimplifier,
artitionFilterOptimizer]}
```

```
grunt> filtered_products 3 = FILTER sales BY NOT (year == 2000);
grunt> DUMP filtered_products 3;
2024-02-13 12:53:52,640 [main] INFO org.apache.pig.tools.pigstats.ScriptState -
res used in the script: FILTER
2024-02-13 12:53:52,641 [main] INFO org.apache.pig.newplan.logical.optimizer.Lc
ptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, DuplicateForEachColumn
```

pRedUtil - Total input paths to

```
(2000,,1000)
(2001,,1500)
(2000,,1200)
(2001,,1500)
(2002,,900)
```